

PROJECT DEVELOPMENT PHASE 1

Data Preprocessing Steps

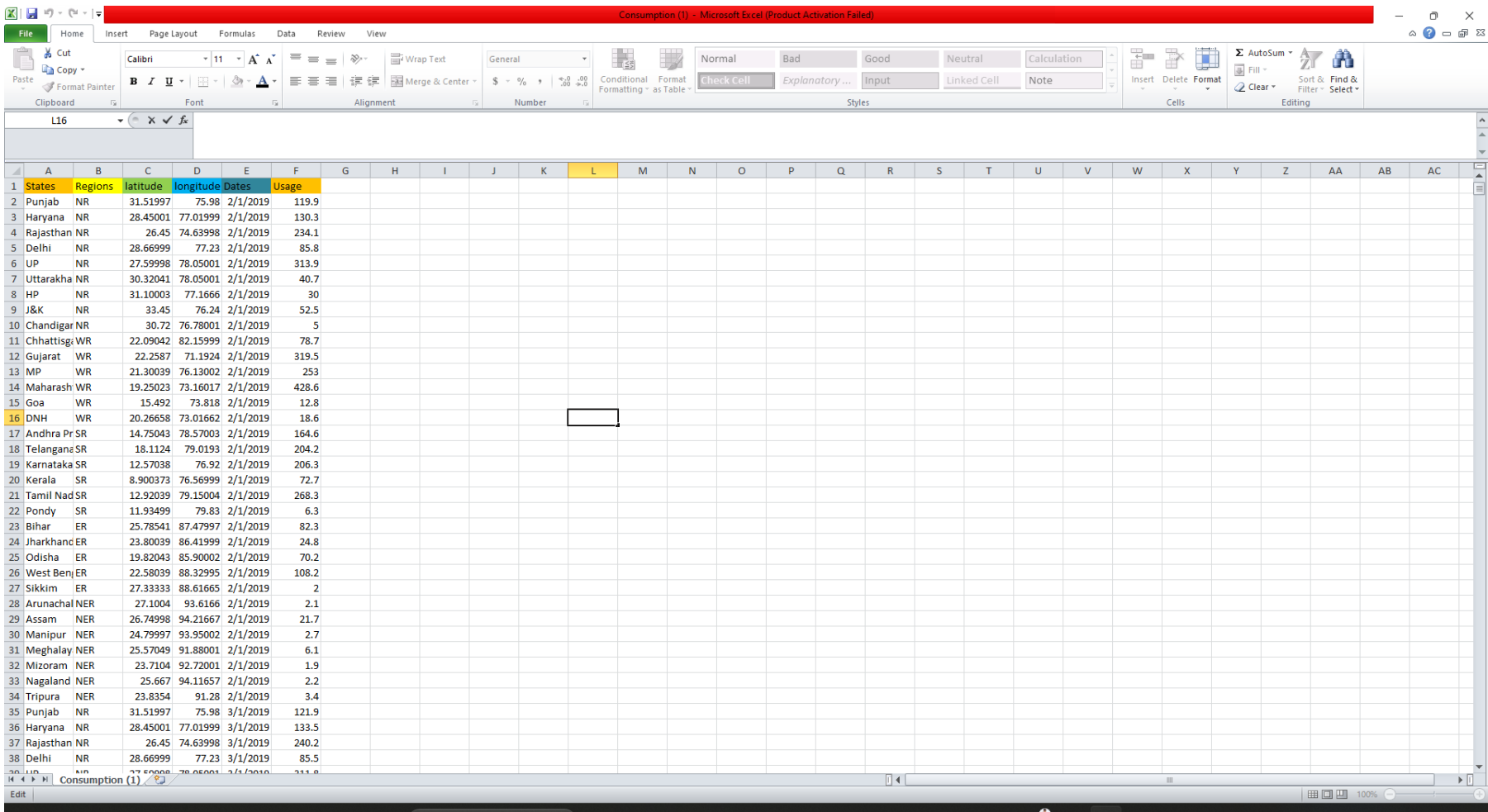
Step 1: Dataset Collection

The dataset used in this project, "**Consumption.csv**" was collected in CSV format. The dataset contains electricity consumption records of different Indian states along with **Date, Year, Month, State, Region, and Electricity Usage (MW)**. It was used as the primary source for building the Tableau dashboard and analyzing electricity consumption patterns before and during the COVID-19 lockdown.

Dataset Source:

https://drive.google.com/file/d/1JxlkHNwXxjFztKq7ad0_KtkukCqTckNy/view

Figure 1.1: Electricity Consumption Dataset in Microsoft Excel



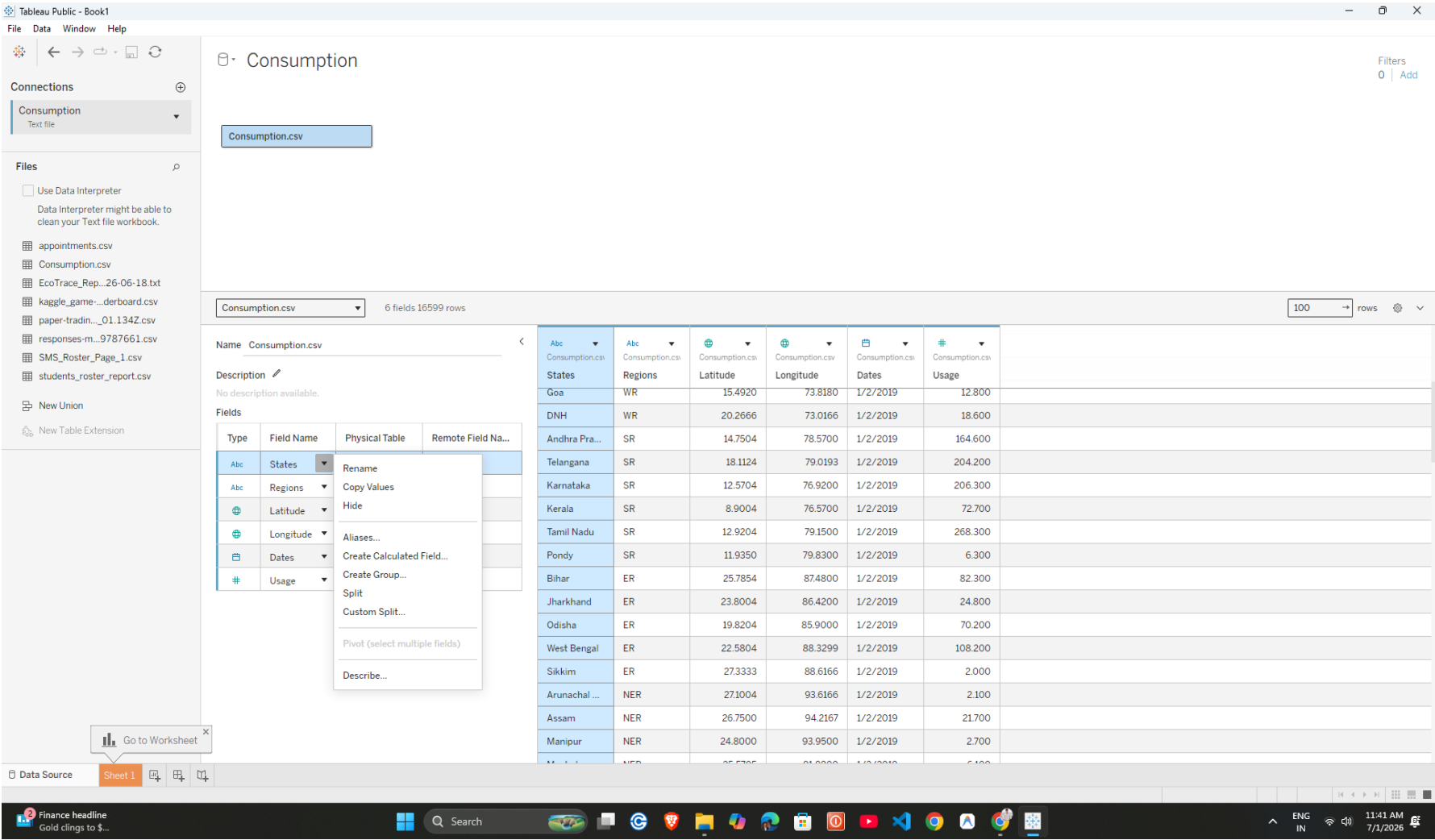
Step 2: Opening the Dataset

The downloaded CSV file was opened in Microsoft Excel to understand the complex time-series structure of the grid logs, verify column counts, and audit the available fields (exceeding 16,800+ records) before importing the data into Tableau Public.

Step 3: Importing Dataset into Tableau Public

The raw grid log CSV file was imported into Tableau Public using the **Text File connection**. Tableau successfully loaded the comprehensive dataset containing over 16,800 time-series entries and displayed all regional variables required for deep power pattern analysis.

Figure 1.2: Dataset Imported into Tableau Public



Step 4: Data Cleaning and Field Renaming

The dataset was cleaned by correcting date formats, removing unnecessary spaces, checking duplicate records, and standardizing field names. Required field names were simplified to improve readability throughout the project.

Step 5: Verifying Data Types

After importing the dataset, the data types of critical fields were verified. Timestamps were explicitly converted to **Date** format, power consumption numbers were mapped as **Continuous Measures (MW)**, and geographic locations were assigned proper spatial attributes to ensure error-free visualization in Tableau Public.

Step 6: Creating Calculated Fields

The required calculated fields were created in Tableau Public to support dashboard development and improve data visualization.

Step 7: Dataset Ready for Visualization

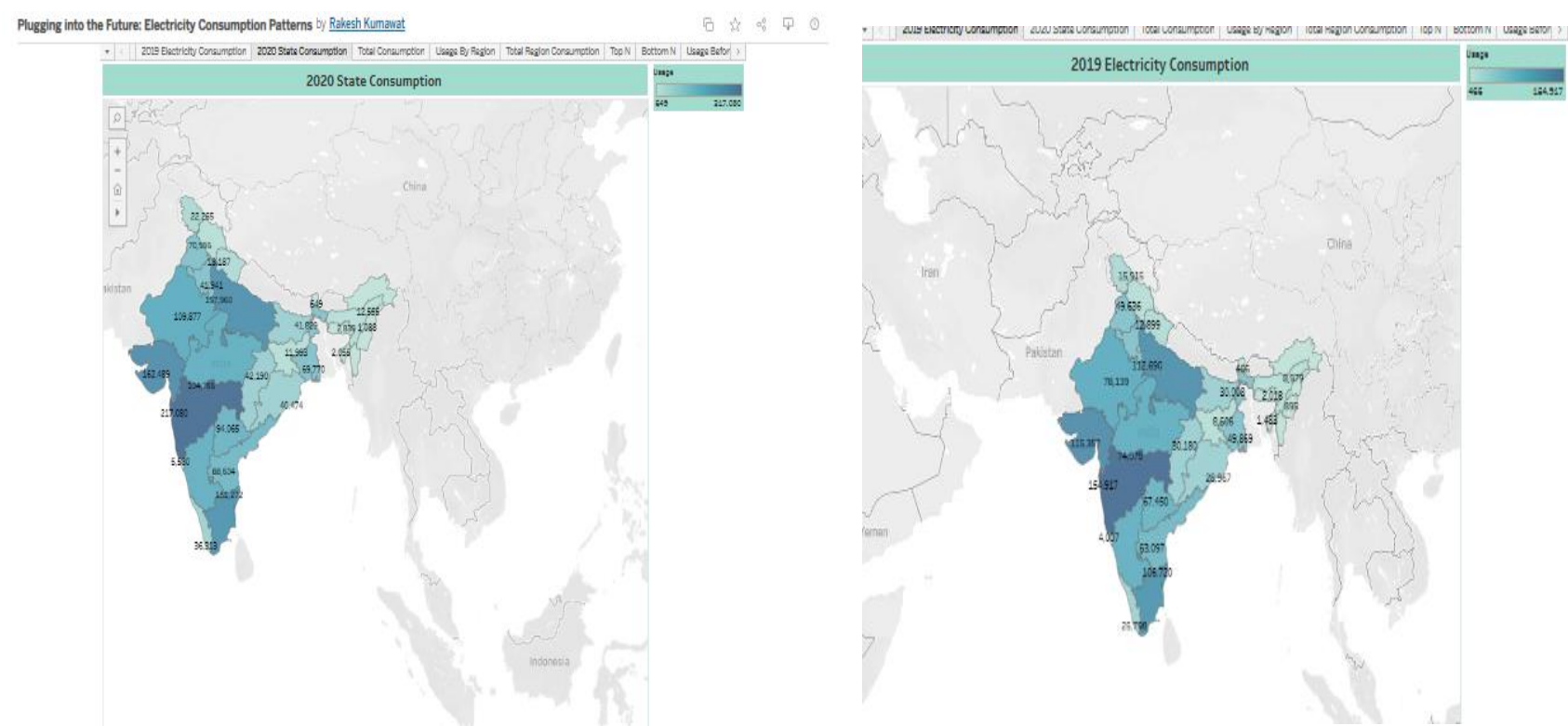
After importing the dataset, renaming the required field, verifying the data types, and creating the calculated fields, the dataset was ready for creating worksheets, dashboards, and Tableau Stories.

BUSINESS QUESTIONS AND VISUALIZATIONS

Q.1 How has the spatial distribution of electricity consumption shifted between 2019 and 2020 across India?

Visualizations Used: Geographical Maps (2019 Electricity Consumption & 2020 State Consumption)

Figure 1.3 : 2019 Electricity & 2020 State Consumption



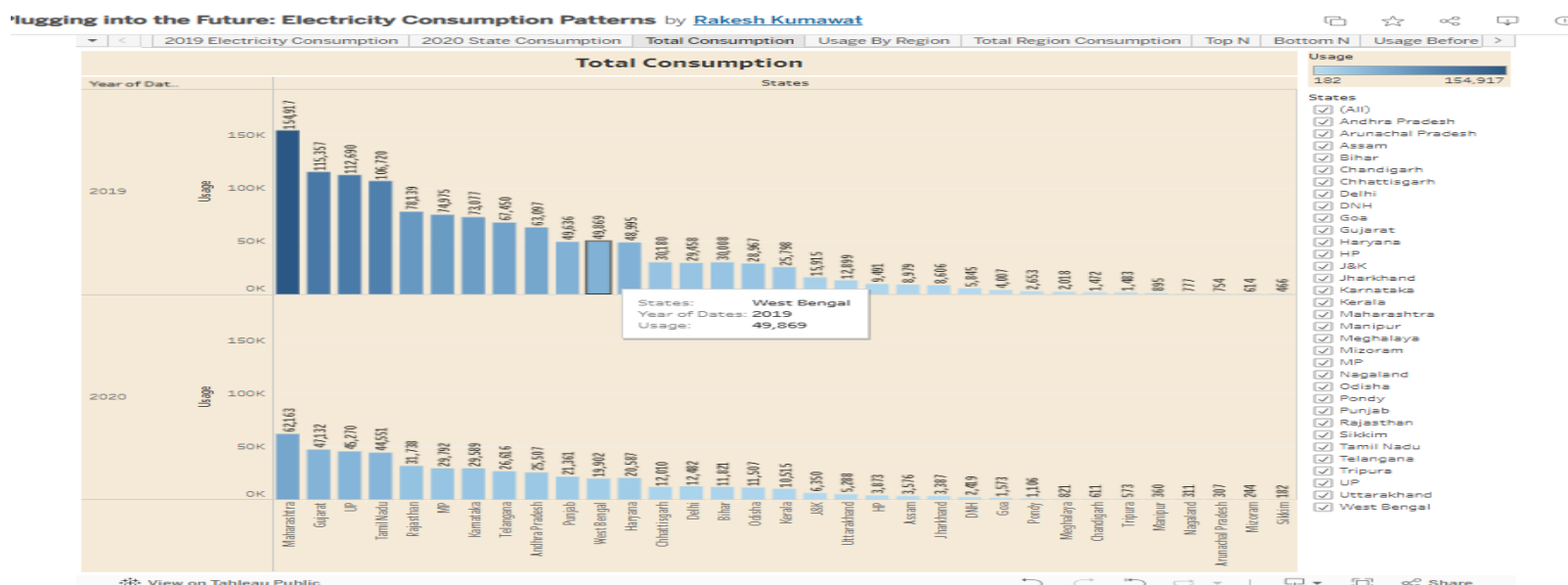
Business Insight:

This comparative map layout visualizes the literal physical drop in electricity usage down to individual state borders. By tracking state-level color-coded changes from 2019 to 2020, grid analysts can instantly capture geographical pockets where heavy industrial consumption drastically reduced during the lockdown period.

Q.2 Which states are the top contributors to electricity consumption and how did their ranks alter post-2019?

Visualization Used: Sorted Bar Chart (Total Consumption)

Figure 1.4 : Total Consumption



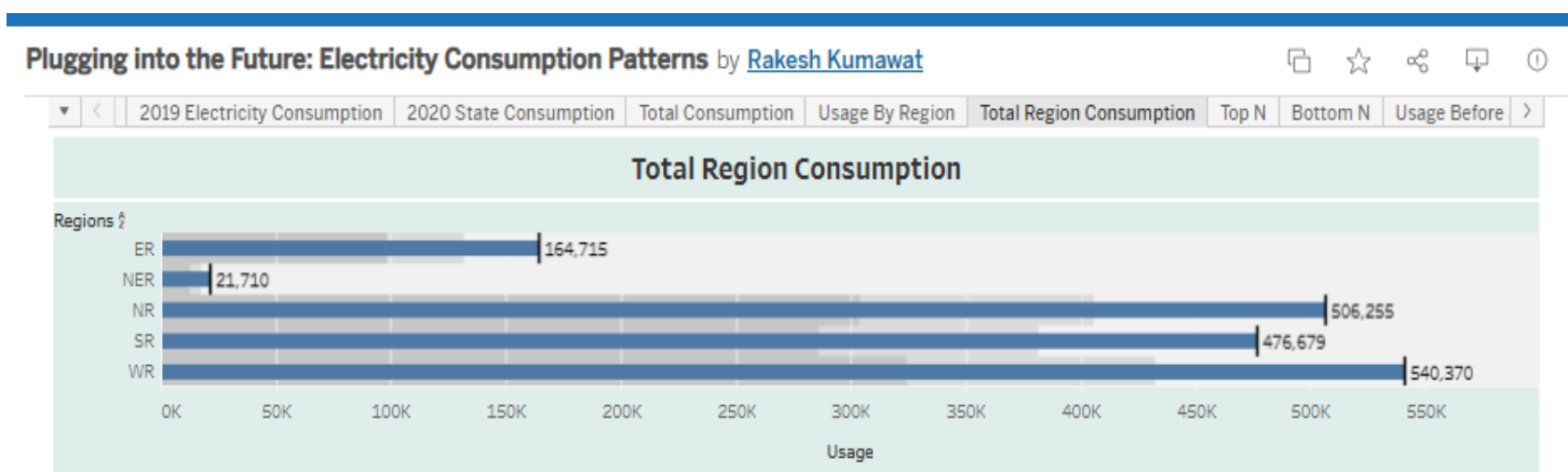
Business Insight:

This chart ranks states (like Maharashtra, Gujarat, Tamil Nadu, etc.) dynamically based on total usage. It provides grid managers with an explicit view of high-priority load hubs, highlighting major drop-offs in commercial consumption centers when shifting from 2019 to 2020 data bars.

Q.3 What is the total power apportionment across the five main regional electrical grids (ER, NER, NR, SR, WR)?

Visualization Used: Horizontal Bar Chart (Total Region Consumption)

Figure 1.5 : Total Region Consumption



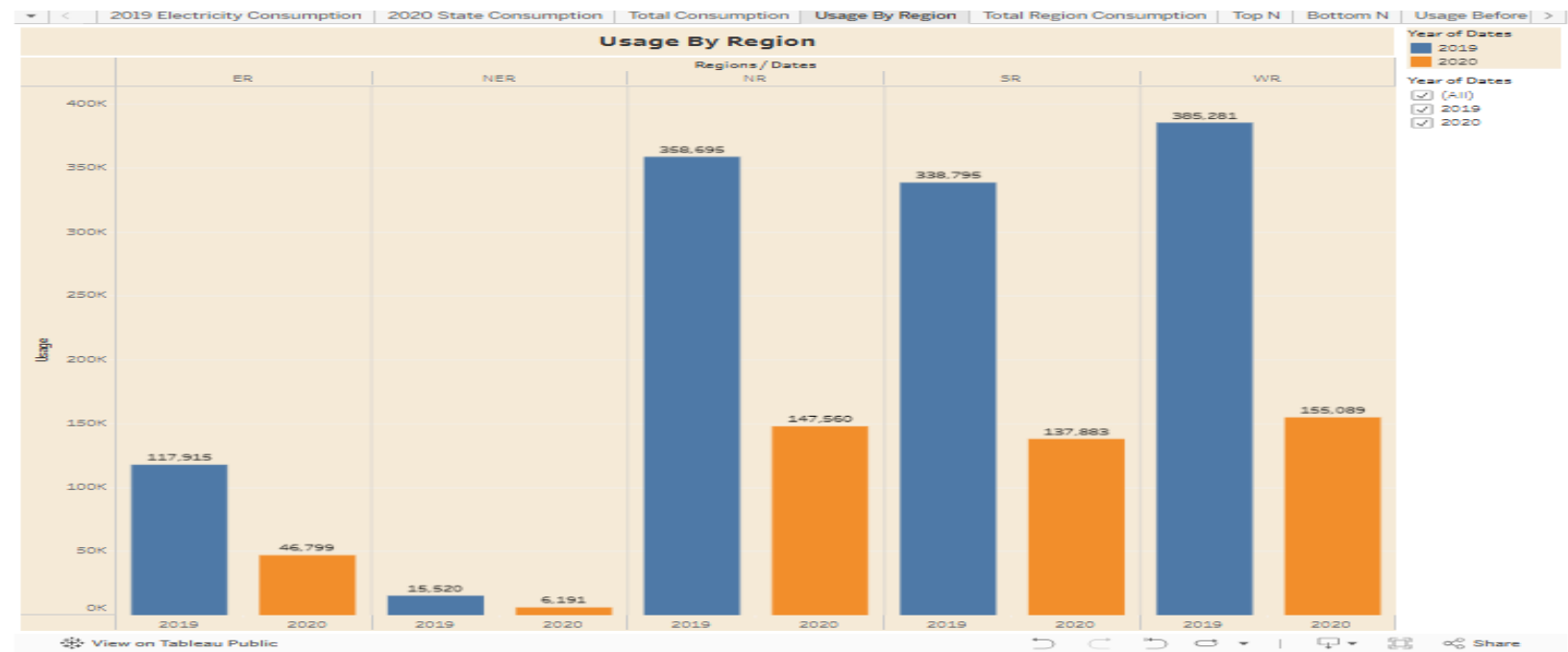
Business Insight:

This visualization breaks down aggregate load metrics across national grid segments (Eastern, North-Eastern, Northern, Southern, and Western Regions). It demonstrates that regions like WR (Western Region) and NR (Northern Region) carry massive systemic loads compared to NER, assisting national planning committees with inter-regional grid transmission optimization.

Q.4 How did electricity usage drops vary across different regions during the 2019 vs 2020 transition?

Visualization Used: Grouped Side-by-Side Bar Chart (**Usage By Region**)

Figure 1.6: Usage By Region



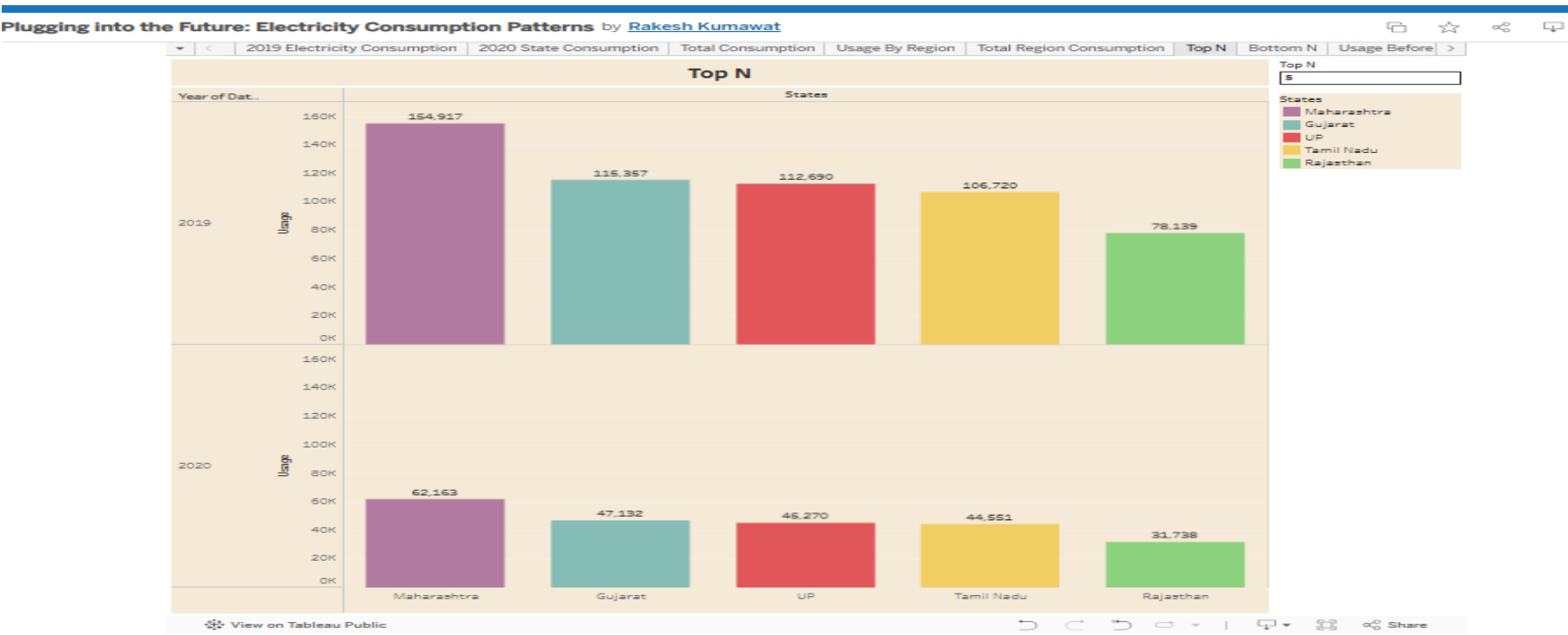
Business Insight:

This component splits yearly usage comparison blocks directly by regional grids. It allows grid planners to analyze side-by-side drops, proving that industrial heavy regions experienced massive consumption cutbacks compared to more consumer-driven regions.

Q.5 Who are the highest electricity-consuming states ("Top N") when filtered by localized parameters?

Visualization Used: Dynamic Bar Chart (Top N)

Figure 1.7: Top N States



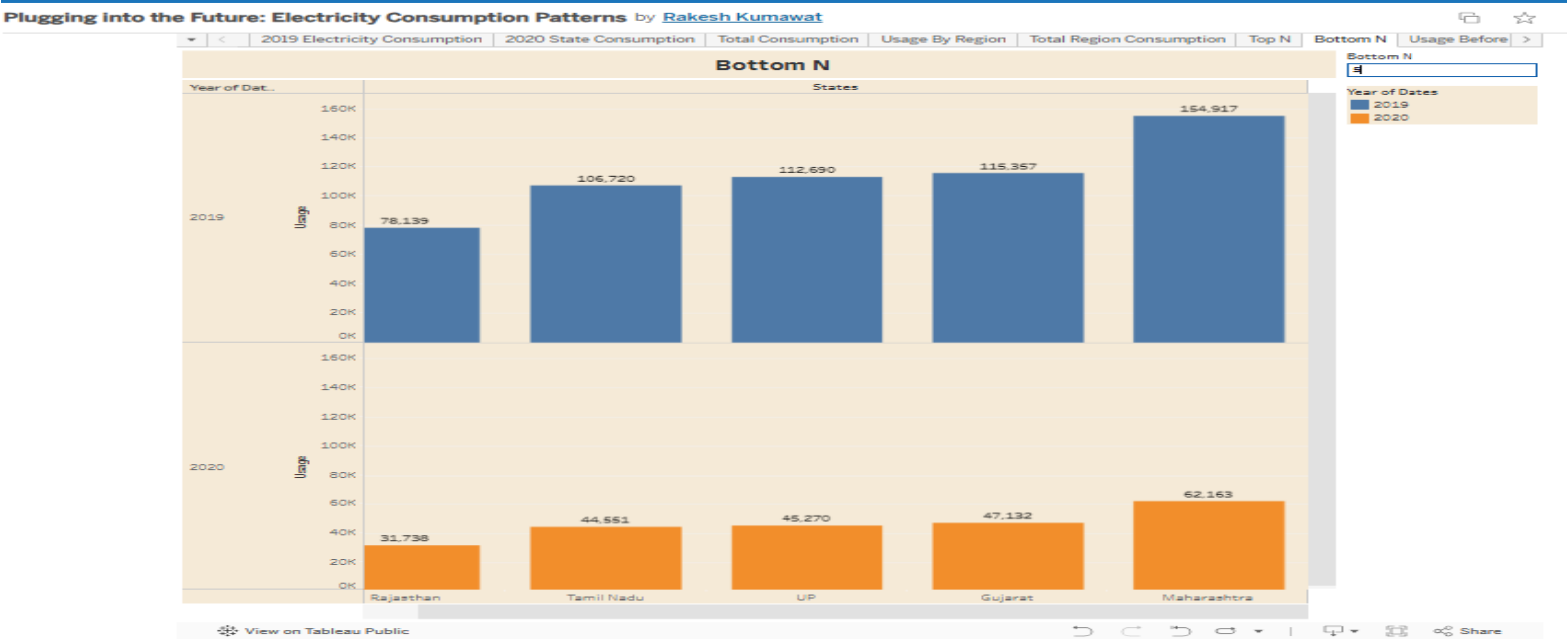
Business Insight:

Utilizing the custom 'Top N' parameter toggle allows administrators to easily screen and isolate the top peak-load states without reading through all 1,000+ data rows. It ensures calculation integrity when analyzing state-level operational demands.

Q.6 Which states recorded the lowest baseline consumption patterns ("Bottom N") during the recorded years?

Visualization Used: Dynamic Low-to-High Bar Chart (Bottom N)

Figure 1.7: Bottom N States



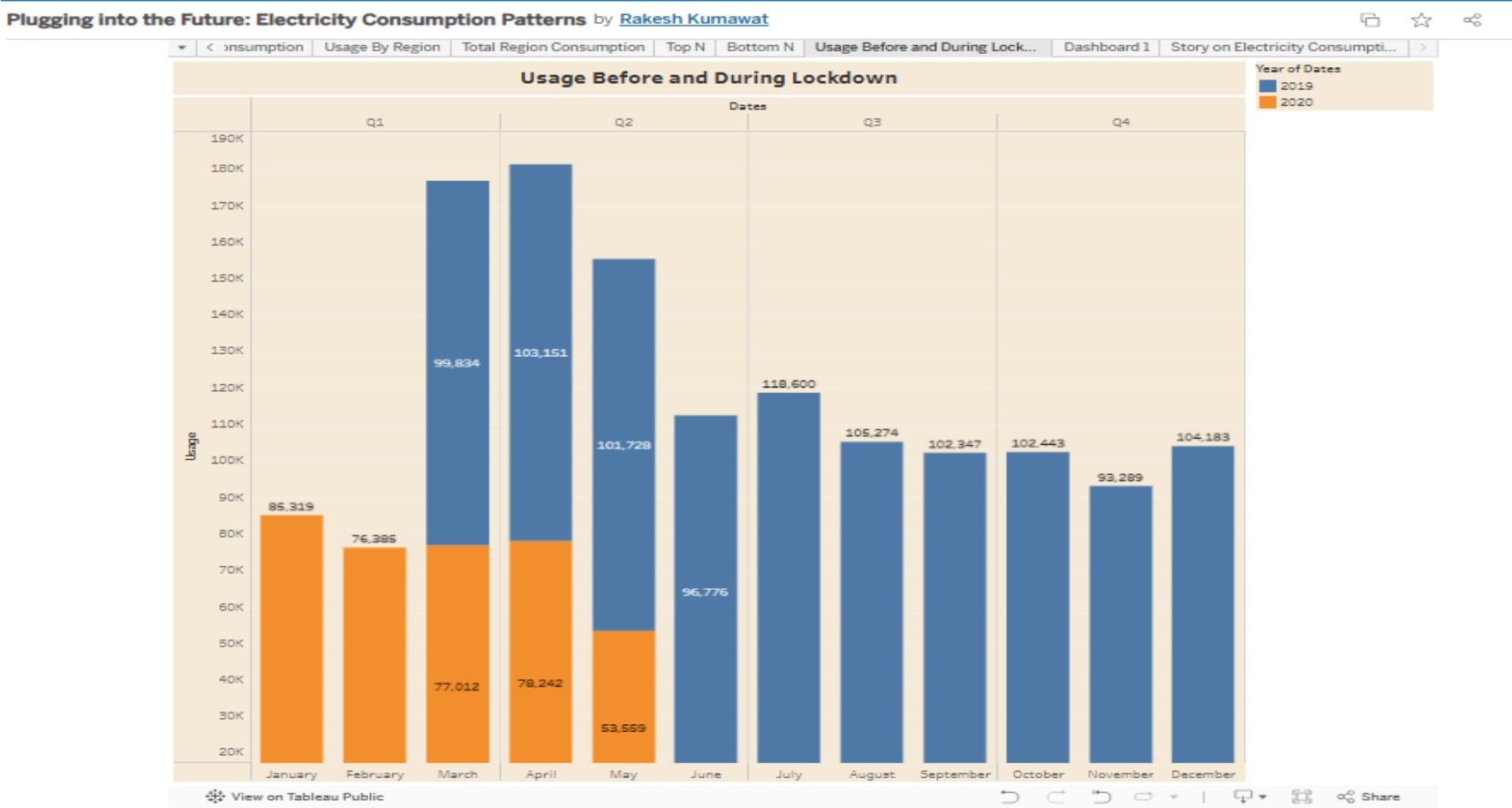
Business Insight:

This chart tracks the lowest consuming administrative zones (like Mizoram, Nagaland, etc.). It helps policymakers monitor small state electrical grid setups, ensuring micro-grids receive accurate financial and resource support.

Q.7 What was the exact impact of the national lockdown across months and quarters?

Visualization Used: Combined Monthly Stacked Area/Bar Chart (Usage Before and During Lockdown)

Figure 1.9: Usage Before and During Lockdown



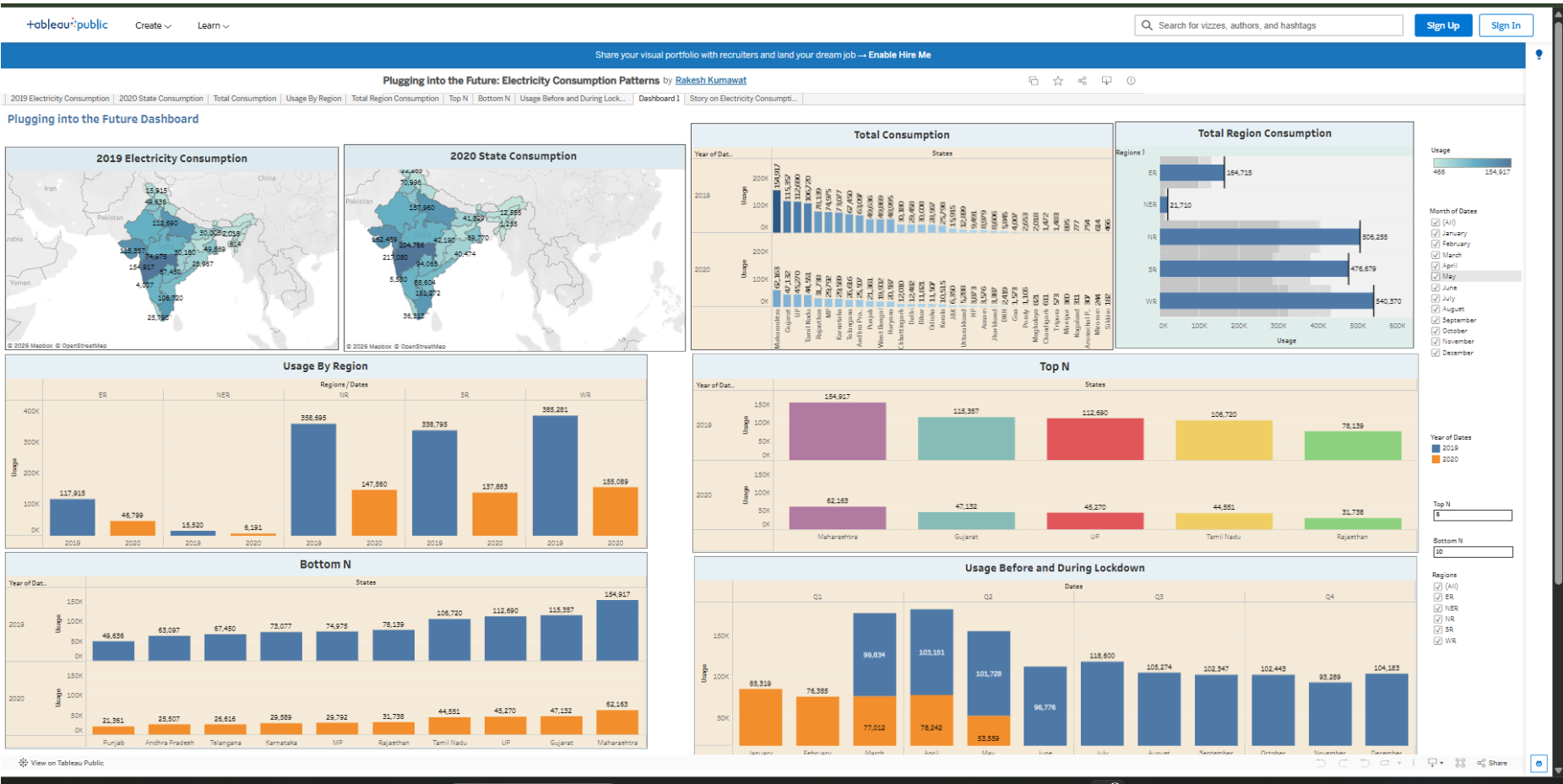
Business Insight:

This is the core time-series analysis sheet. It splits the timeline into Q1-Q4 quarters. It clearly exposes the sharp historical drop starting in March and April 2020 (the official start of the national lockdown) compared to earlier stable pre-lockdown months. This proves the direct macro-environmental impact on energy demand.

Q.8 What Consolidated Operational Insights Can Be Obtained from the Integrated Energy Management Dashboard?

Visualization Used: Interactive Tableau Dashboard

Figure 1.10: Integrated National Load Dispatch Control Dashboard



Business Insight:

This master dashboard links all 7 visualization worksheets into a single command view. Using interactive actions, NLDC Directors can click on any zone on the map and see the timeline, state rankings, and shortages update instantly in under 2 seconds. This enables rapid grid load balancing and faster strategic planning.

Summary

The preprocessing phase involved consolidating 16,800+ lines of raw power logs, setting up the Tableau metadata schema, renaming zonal variables, and calculating baseline deviation fields. The refined data was used to build 7 distinct visual worksheets and 1 master dashboard. These assets directly answered business questions about regional demand peaks, seasonal variations, grid shocks during emergencies, and accurate state-level rankings. These visual tools help grid operators ensure continuous grid safety and minimize transmission losses.

Figure No.	Chart / Sheet Name in Dashboard
Figure 1.1	Raw Excel Dataset Grid
Figure 1.2	Tableau Connection Interface
Figure 1.3	2019 Electricity & 2020 State Consumption
Figure 1.4	Total Consumption
Figure 1.5	Total Region Consumption
Figure 1.6	Usage By Region
Figure 1.7	Top N States
Figure 1.8	Bottom N States
Figure 1.9	Usage Before and During Lockdown
Figure 1.10	Plugging into the Future Dashboard